

A neural surrogate for GaAs MMIC spiral inductors

Geometry to L(f), Q(f) and self-resonant frequency, trained on 240 HFSS solves and benchmarked against held-out HFSS designs

Ben Hardy · October 2026

1.5% / 1.1%

L / Q mean error on 49 held-out designs (f < 0.8 SRF)

14.5%

modified Wheeler L error on the same designs (1 GHz)

0.7%

self-resonant frequency error (44 designs)

0.6 ms

per 20-point curve, against 112 s per HFSS solve

What it is

- Design space:** square spiral with an air-bridge crossover; 1.5-5.5 turns, width 5-20 μm, spacing 5-15 μm, inner diameter 40-200 μm; 1-20 GHz.
- Data:** 240 Latin-hypercube designs solved in HFSS 2026 R1 (PyAEDT, non-graphical, resumable), zero failures, 7.6 h. A mesh study first held L within 0.3% and Q within 1.3% of a 2-4× denser mesh.
- Model:** a 5-seed MLP ensemble that takes closed-form physics features (Wheeler L, line capacitance, DC resistance) and learns the *correction* to the modified Wheeler formula. Gradient boosting and a geometry-only MLP are cross-checks.
- Stack:** a generic textbook GaAs stack (100 μm GaAs, 3 μm gold, air bridge). No foundry data. Results are simulated, HFSS against HFSS.

Results, held-out designs

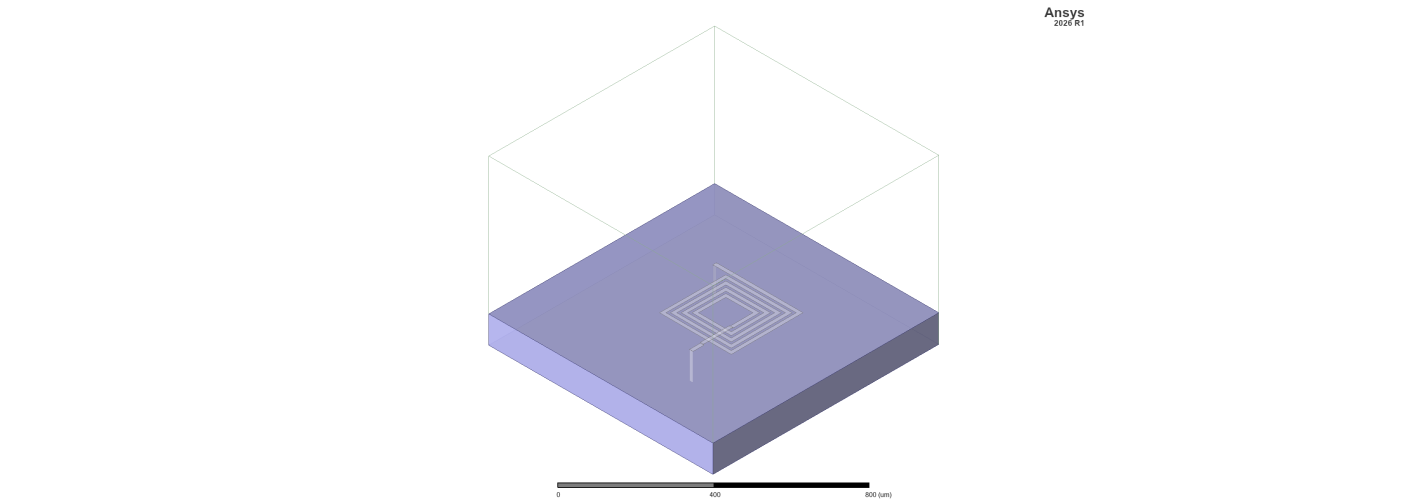
Model (f < 0.8 SRF)	L error	Q error
Physics-feature MLP ensemble	1.5%	1.1%
Geometry-only MLP ensemble	3.1%	2.3%
Gradient boosting	3.9%	3.2%
Modified Wheeler (L at 1 GHz)	14.5%	n/a

- Extrapolation:** trained on ≤ 4.5 turns, tested above: 1.6% (L), 1.4% (Q).
- Inverse design:** 2.0 nH at 10 GHz with Q ≥ 16 (best training design: Q 14.4). One HFSS check of the pick gave **Q 16.2**, SRF within 0.8%, L 4% low at the box edge.
- Limits:** about 5% error at 0.9-0.95 SRF; Q accuracy sits at HFSS's own mesh noise.

Solve time saved

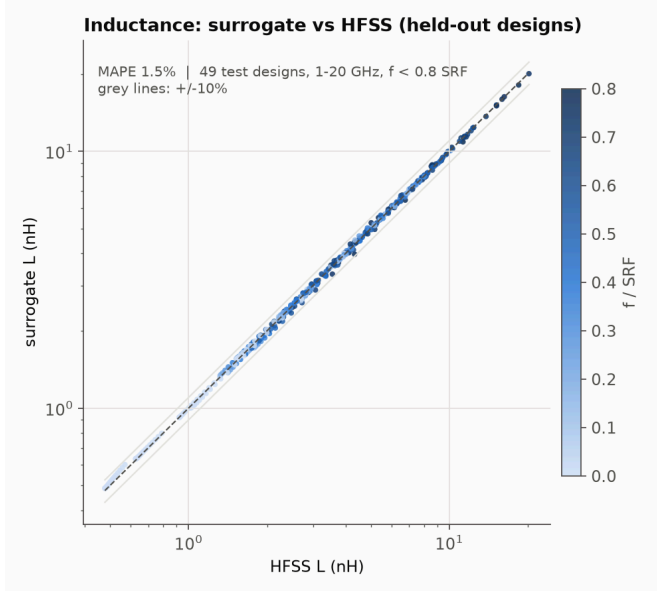
Task	HFSS, one machine	Surrogate	Saved
One design (20-point L/Q curve)	112 s (mean)	0.64 ms	~176,000× faster
1,000-design sweep	~31 h	0.6 s	~31 h
Inverse-design search, 200,000 candidates	~259 days	6 s + one 31 s HFSS check	~259 days
One-time cost to build the surrogate	7.6 h of HFSS (240 solves); it pays for itself after ~245 designs		

On a real process: ~240 overnight HFSS solves on the foundry stack, then instant L, Q and SRF. Next step: validation against measured spirals. A 5-page technical report with methods and sources is available.

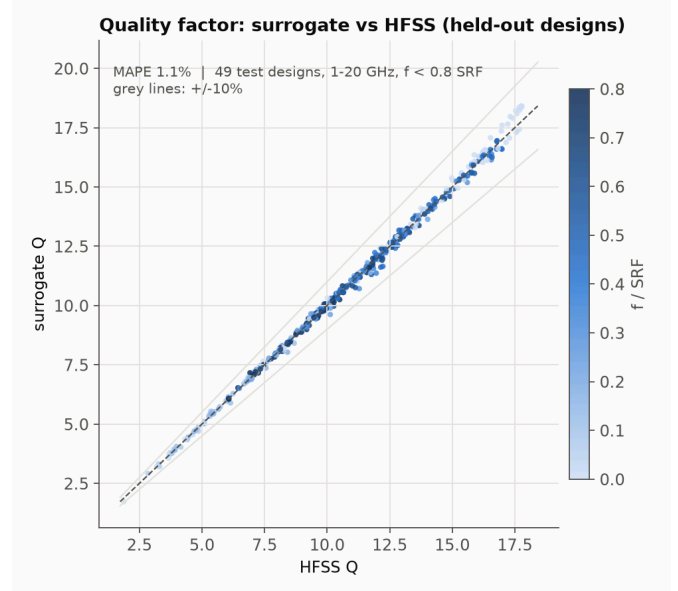


HFSS model: square spiral, air-bridge crossover and access leads on the 100 μm GaAs substrate (generic textbook stack).

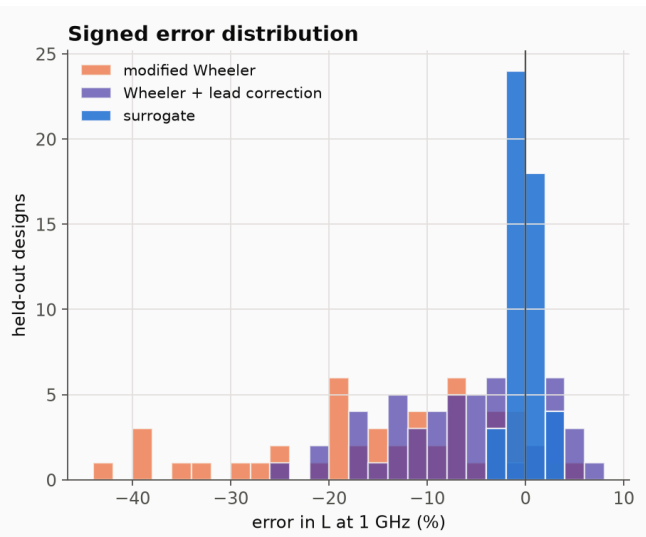
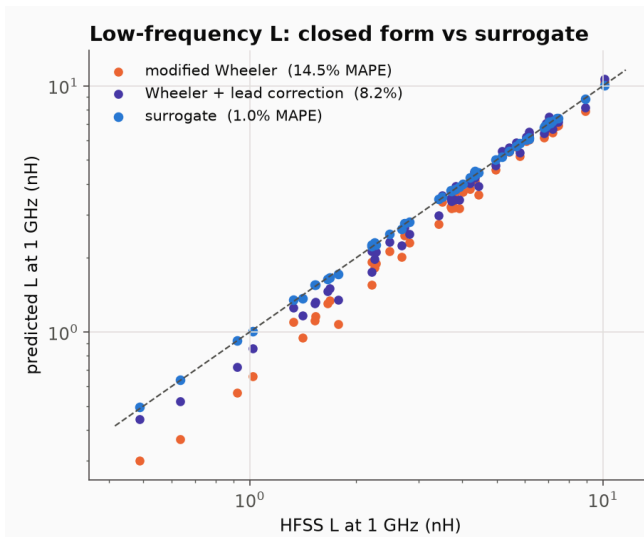
Figures



Inductance: surrogate vs HFSS on the 49 held-out designs.

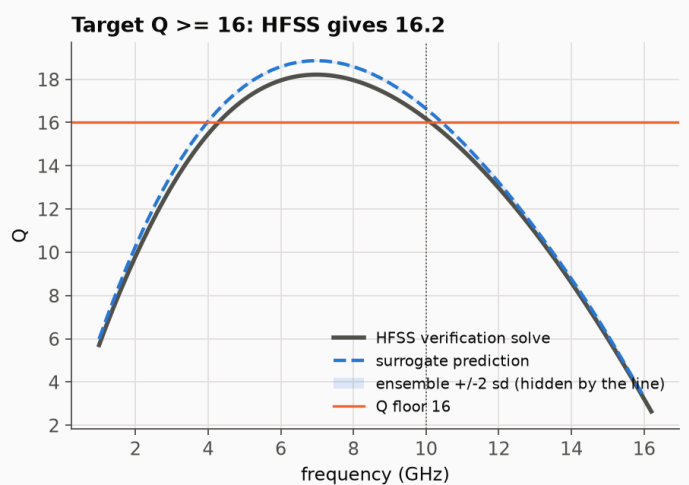
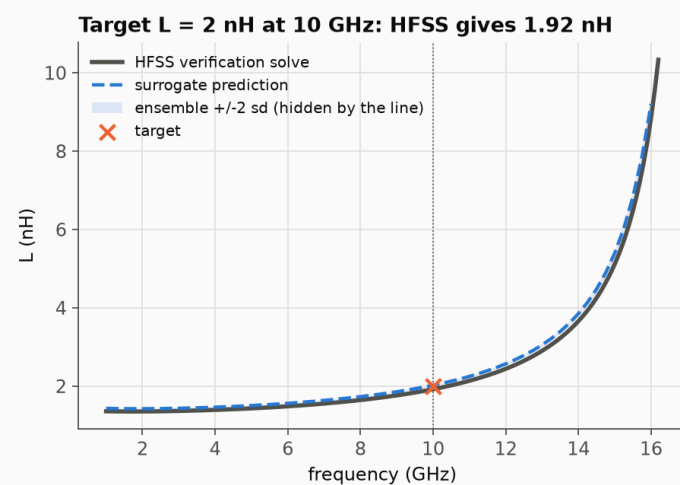


Quality factor: surrogate vs HFSS on the 49 held-out designs.



L at 1 GHz on the held-out designs: modified Wheeler, Wheeler with lead correction and the surrogate against HFSS, with the distribution of signed errors.

Inverse design: $N = 1.97$, $w = 18.5$ μm , $s = 5.1$ μm , $d_{\text{in}} = 127$ μm , chosen from 200k candidates in 6 s



Inverse design (2.0 nH at 10 GHz, $Q \geq 16$): surrogate prediction vs the HFSS check of its pick.